PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH
BADJI MOKHTAR UNIVERSITY -ANNABA-
FACULTY OF MEDICINE

DEPARTMENT OF MEDICINE
LABORATORY OF MEDICAL AND SURGICAL ANATOMY

SKELETAL SYSTEM

Prepared by Professor BOUKOFFA SACIA
Professor of General Anatomy

Academic Year 2025-2026

SKELETAL SYSTEM

Plan

Introduction

1. Roles of the skeleton
2. Cartilaginous part of the skeleton
3. Bony part of the skeleton

3.1. Definition
3.2. Skeletal Structure
3.3. Classification of bones
3.4. Bone Morphology
3.5. Bone Structure
3.6. Bone Growth
3.7. Radiological Anatomy

Conclusion

Goals

1. Understanding the human skeleton.
2. Knowing the different types of bones.
3. Understanding bone morphology.
4. Understanding the structure of bone.

SKELETAL SYSTEM

Introduction

The skeletal system or skeleton of the human body is represented by all the bones and cartilages that protect the organs, allow movement and constitute a framework on which the muscles are arranged.
In humans, the skeleton is an endoskeleton comprising both cartilaginous and bony parts. The study of the skeletal system constitutes osteology.

1. Roles of the skeleton

The bony skeleton performs the following functions:

- ➢ The skeleton forms the framework that supports the body.
- ➢ It protects the organs.
- ➢ Bones represent the site of muscle insertions.
- ➢ Bones store mineral salts (Ca+).
- ➢ Bones are responsible for the production of blood cells.
- ➢ Cartilage facilitates movement and protects the ends of bones from wear and tear

2. Cartilaginous part of the skeleton

The cartilaginous part of the skeleton is made up of a strong but elastic tissue, devoid of blood vessels and nerves : cartilage. In the fetus, the majority of the skeleton is cartilaginous, whereas in the adult, some cartilaginous structures persist.

2.1.Cartilage classification

2.1.1. Hyaline cartilage : translucent, bluish and vitreous. Examples include articular cartilage, the thyroid and cricoid cartilages of the larynx, the tracheobronchial tree and nasal cartilages.

2.1.2. Fibrocartilage : whitish and resistant, rich in collagen fibers. Examples include the intervertebral disc, the meniscus of the knee, and the acetabular labrum.

2.1.3. Elastic cartilage : yellowish and rich in elastic fibers. Examples include the auricular cartilage, the epiglottis, and the small cartilages of the larynx.

3. Bony part of the skeleton

3.1.Definition

The bony skeleton is made up of bones, which are rigid and hard structures of the human body. There are 206 bones; this number varies due to the presence of supernumerary bones (at the extremities and in the skull). These supernumerary bones are the sesamoid bones and the Wormian bones.

3.2.skeletal structure

The bony skeleton is divided into two parts :

- ➢ The axial skeleton is formed by the spine, the bony head, the rib cage, and the hyoid bone.
- ➢ The appendicular skeleton is symmetrical, attaches to the axial skeleton; it is formed by the bones of the thoracic limbs and the bones of the pelvic limbs.

3.3.Bone classification

Bones are classified, according to their three dimensions, into three types :

- ➢ Long bones

These are bo ates over their width and thickness; it has three parts:

Yellow: axial skeleton
Red: shoulder girdle
Blue: pelvic girdle
Move and green: appendicular skeleton

diaphysis (body).

imal and distal epiphyses, which bear the articular

These are bones whose length and width predominate over their thickness; it has: two faces, edges and angles.

- ➢ Short bones

These are bones whose the three dimensions are almost equal.

- ➢ Irregular bones, for example, the vertebrae.

3.4.Morphology

Bones have an irregular surface with protrusions (eminences, reliefs) and depressions (cavities).
The bony protrusions are oftwo types :

- ➢ The articular eminences or head, for example the humeral head.
- ➢ Non-articular eminences : They are used for muscle or ligament insertions and can be :
  - The process (apophysis) is a large eminence of variable shape, for example, the coracoid process.
  - The tuberosity is a rounded eminence, for example, the major tuberosity of the femur.
  - The tubercle is a small, rounded eminence, the infra-glenoid tubercle.
  - The acute eminence of the sciatic spine of the iliac bone
  - The linear eminence ridge, for example, the iliac crest.

Bone cavities are of two types :

- ➢ The articular cavities correspond to the articular eminences, for example the trochlear notch of the ulna.
- ➢ Non-articular cavities can be:

  - The groove (channel) is the place where a tendon, nerve or vessel passes, for example the radial groove of the humerus.
  - A dimple is a hollow that is more or less wide and deep, for example the olecranon fossa of the ulna.
  - The foramen (hole) is an opening in the bone, for example, an obturator foramen.
  - The incision (notch) is a cavity dug at the level of an edge, for example a sciatic notch.

3.5.Bone structure

Bone is a specialized, dense and strong connective tissue called osseous tissue.it is :

- Rich in phosphate and calcium carbonate
- Richly vascularized
- Constantly evolving
- Formed by 3 types of bone cells contained within a calcified intercellular matrix.

Bone presents itself from the surface to the depth :

➢ The periosteum is a whitish fibrous connective-elastic membrane that covers the bone except for the articular surfaces.

➢ Compact bone : is a hard and dense substance, comprising :

- Osteons are the histo-physiological unit consisting of a vascular canal and concentric bony lamellae that surround it.
- The arcuate lamellae interposed between the osteons.

The compact bone forms the boundary of the medullary canal in long bones. This canal contains the bone marrow.

➢ Spongy bone: is made up of trabeculae or bony lamellae delimiting small cavities. Located at the level of the epiphyses, it forms the intermediate layer of flat and short bones.
At the level of the bones of the vault of the skull it is called diploe and the two layers of compact bone that surround it constitute the external and internal tables.

3.6.Bone growth

Ossification begins at the end of the embryonic period.
It is preceded by a cartilaginous or membranous matrix which will be replaced by bone tissue, hence two types of ossification : membrane ossification and endochondral ossification.
Bone growth occurs in two ways:

➢ Growth in length depends on the growth plate located at the diaphyseal-epiphyseal junction.
➢ The growth in width results from the action of the periosteum.

3.7.Radiological Anatomy

The bony skeleton is explored by standard radiography (traumatic pathology especially fractures). Computed tomography and magnetic resonance imaging explore inflammatory and tumoral pathology of the human skeleton.

Conclusion

The skeletal system or skeleton of the human body is represented by all the bones and cartilages that protect the organs, allow movement and constitute a framework on which the muscles are arranged.
In humans, the skeleton is an endoskeleton which includes a cartilaginous part and a bony part.

Bibliography

1. Fernand de Peretti. Manual of General Anatomy. Publisher Ellipses marketing 2010, 331 Pages.
2. Kamina P. Clinical Anatomy Volume 1. General Anatomy. Limbs. Maloine Edition, 1997, 287 Pages.
3. Vincent Delmas,Dominique Brémond-GignacGeneral Anatomy.Elsevier Masson, 2008 - 323 pages

Pr. S. BOUKOFFA
Laboratoire d'Anatomie
Médico-Chirurgicale
Faculté de Médecine – Annaba
boukofas@yahoo.fr







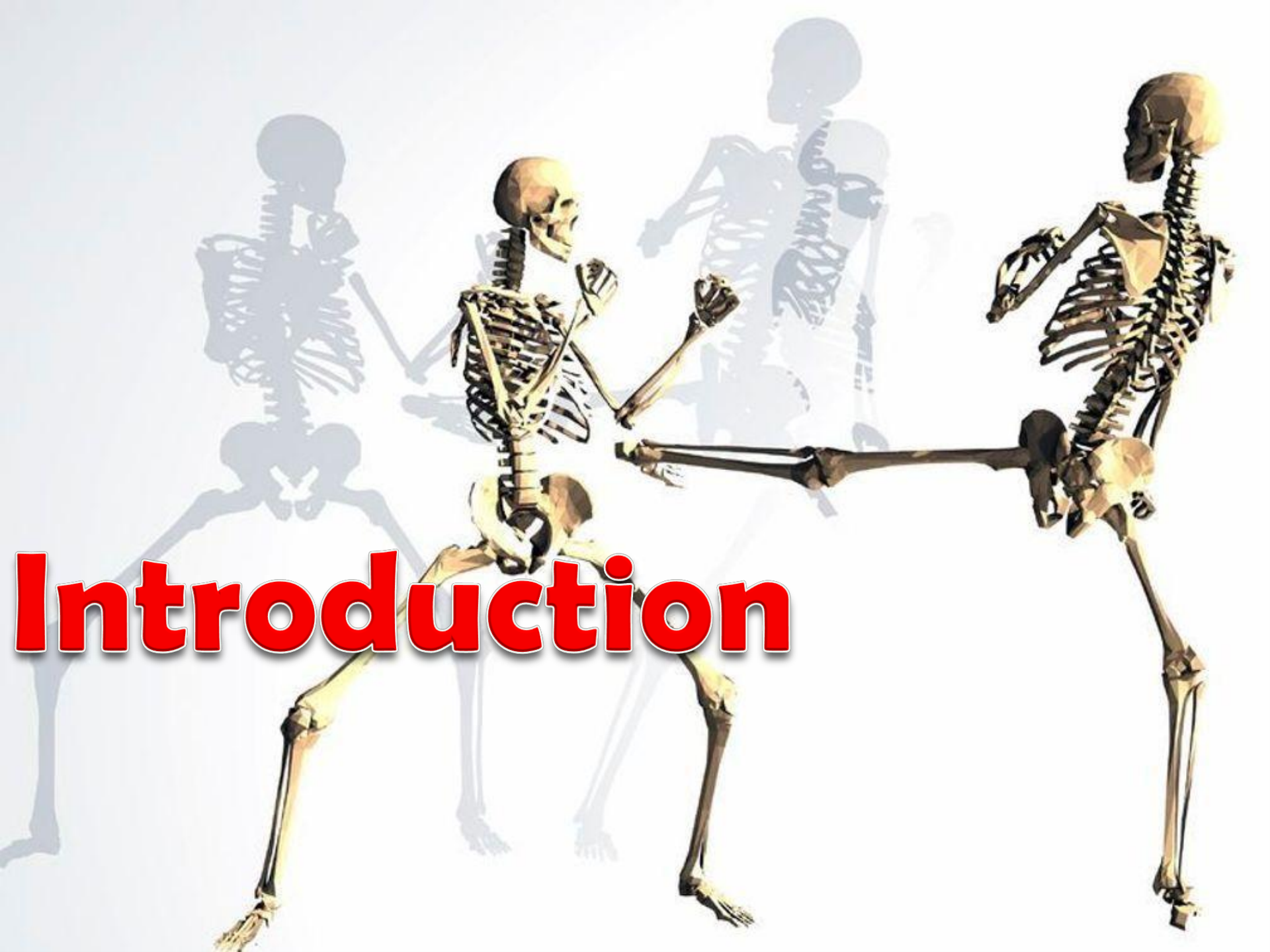
Système squelettique



Plan du cours

- Introduction
- Rôles du squelette
- Partie cartilagineuse
- Partie osseuse
  - Définition
  - Constitution du squelette
  - Classification des os
  - Morphologie des os
  - Structure des os
  - Croissance de l'os

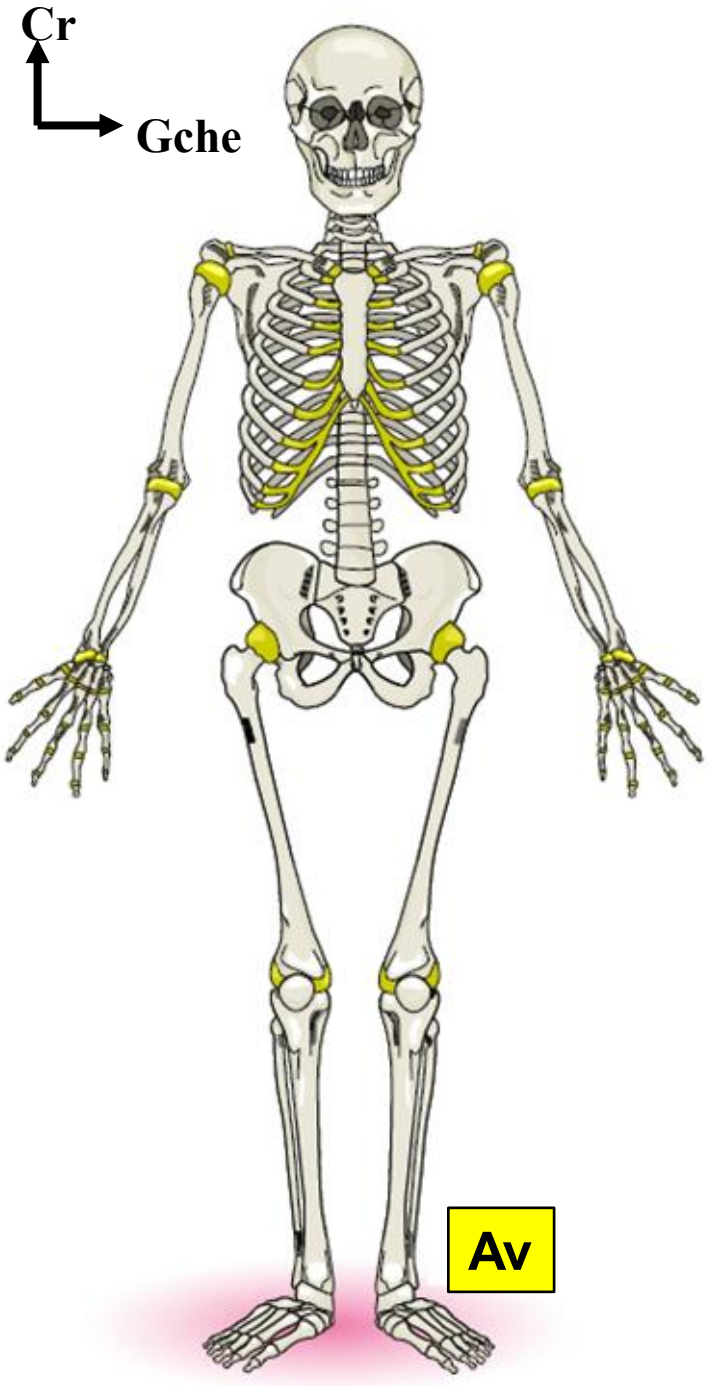



Introduction

Le système squelettique ou **squelette** du corps humain est représenté par l'ensemble des **os** et des **cartilages**.

Chez l'homme le squelette est un **endosquelette** qui comprend une partie cartilagineuse et une partie osseuse.

L'étude du système squelettique constitue l'**ostéologie**.



Rôles du système squelettique



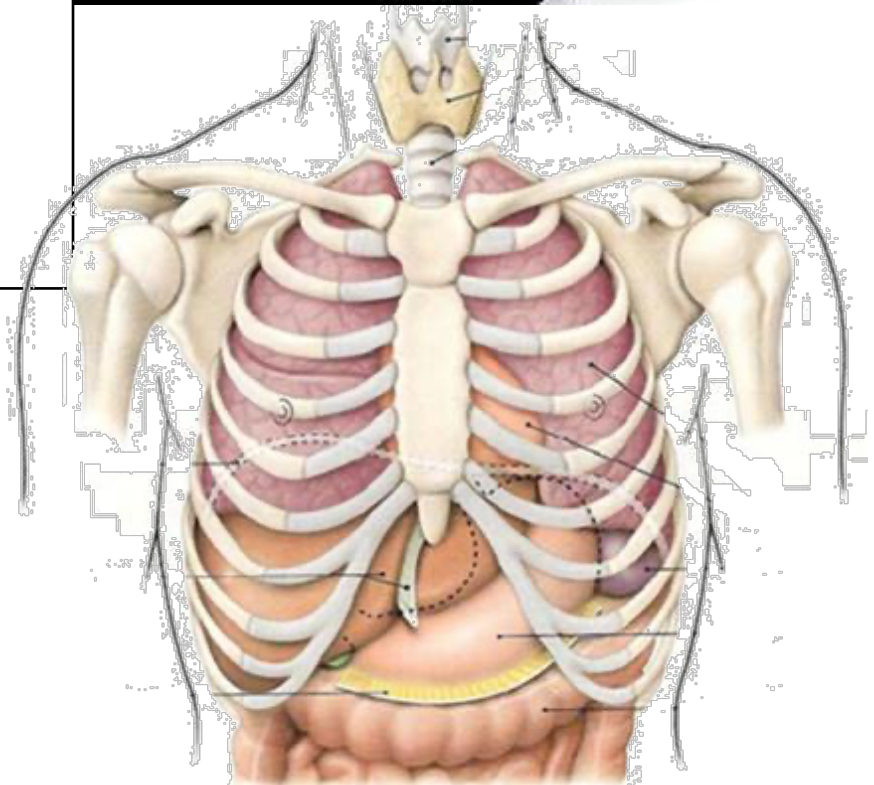
(1)

Le squelette forme la charpente qui soutient le corps



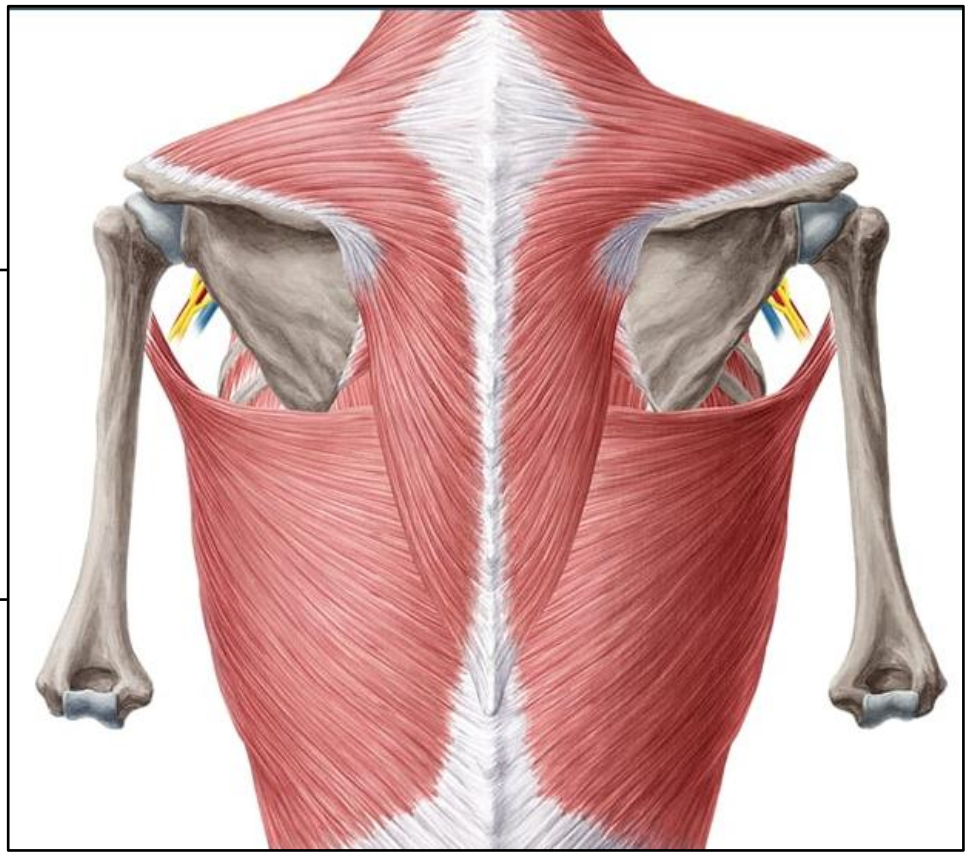
(2) **Le squelette protègent** les organes





(3)

Sites
d'insertions
musculaires



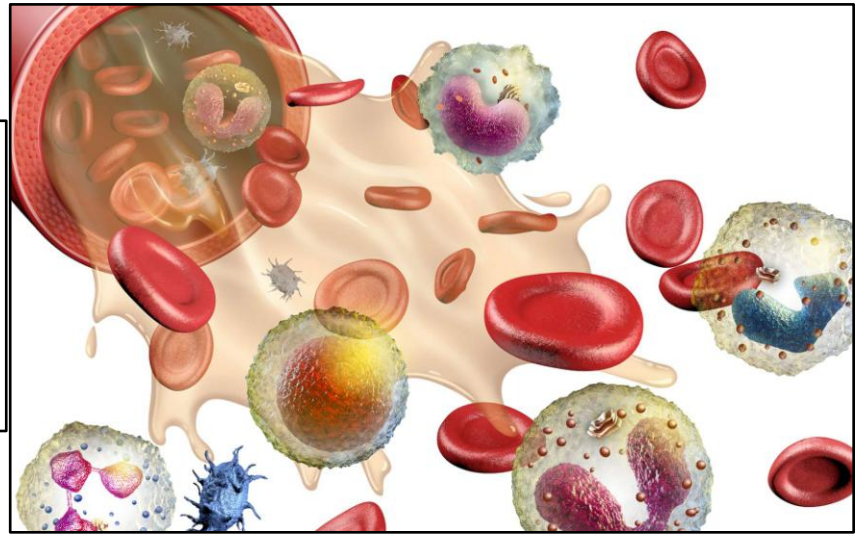
(4)

L'os stockent les sels minéraux (Ca)



(5)

L'os produit les éléments figurés du sang



(6)

Le cartilage facilite **les mouvements et protège** les extrémités osseuses **de l'usure**







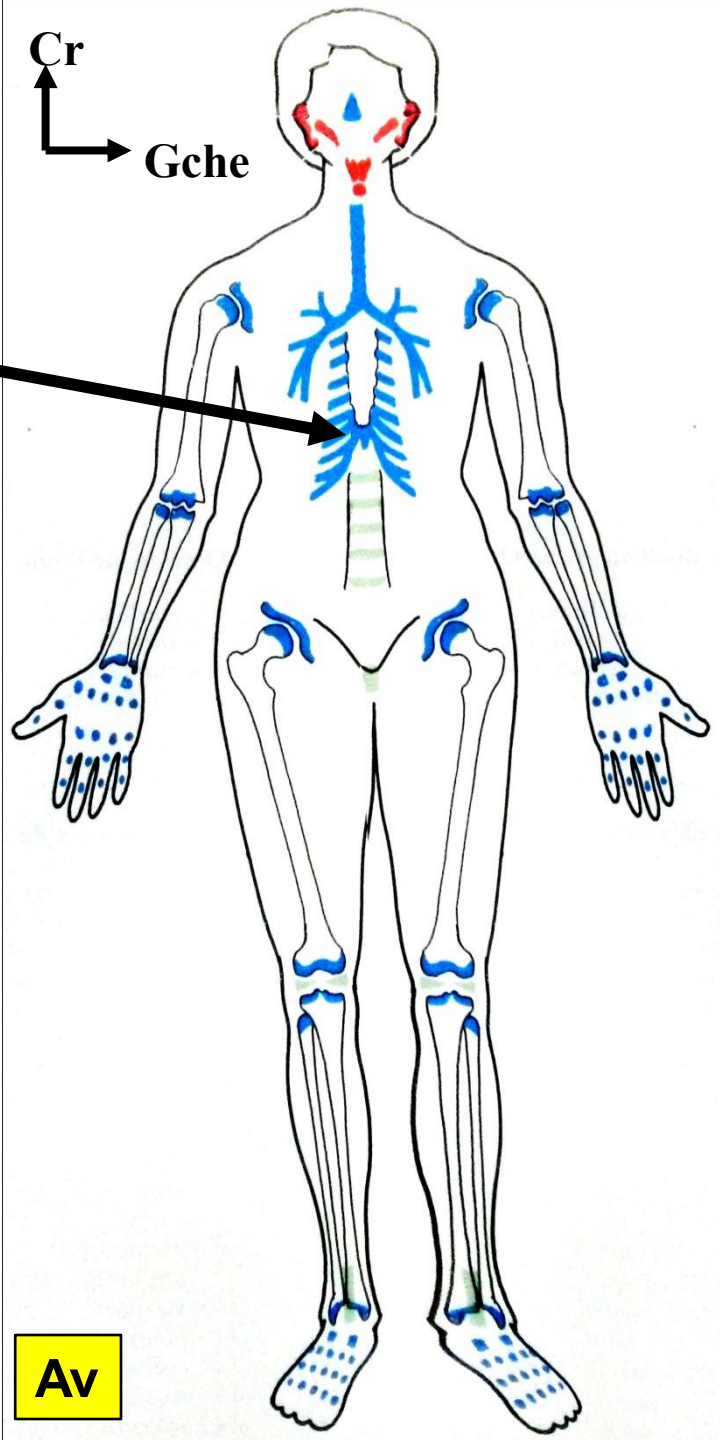
Partie cartilagineuse

Définition

Le **cartilage** **est** un tissu résistant mais élastique, dépourvu de vaisseaux et de nerfs.

Chez le **fœtus** la majeure partie du squelette est **cartilagineuse**.

Chez **l'adulte**, persistent quelques structures cartilagineuses.



Classification

(1) Le cartilage hyalin

Translucide, bleuâtre et **vitreux**

Le larynx(cartilages thyroïde et cricoïde) et l'arbre trachéo-bronchique

Le cartilage articulaire





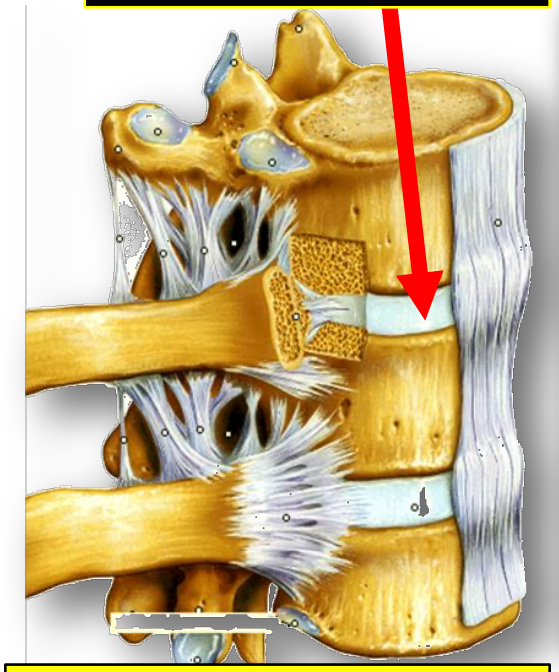
Cartilages nasaux



(2) Cartilage fibreux (fibrocartilage)

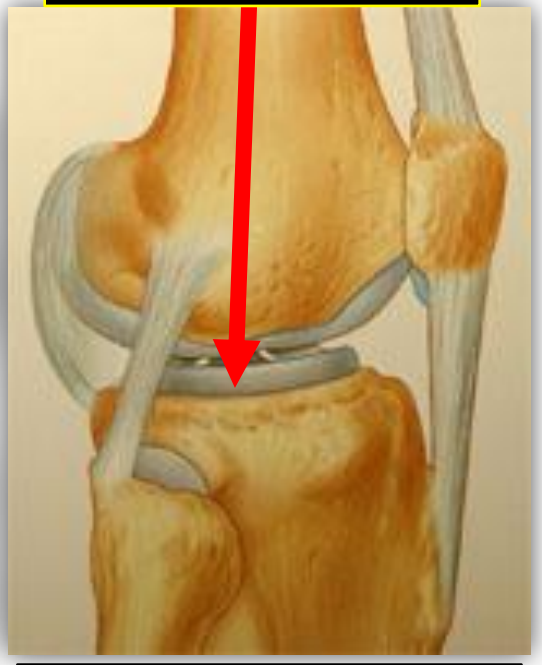
Blanchâtre et **résistant**
Riche en **fibres collagènes**

Disque intervertébral



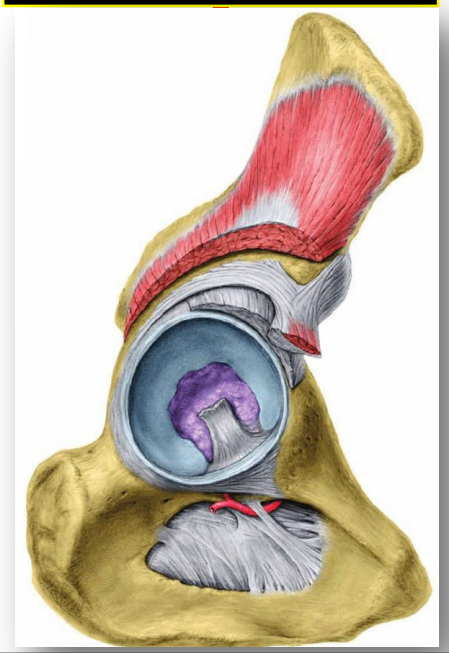
Vue de profil du rachis

Ménisque du genou



Vue de profil du genou

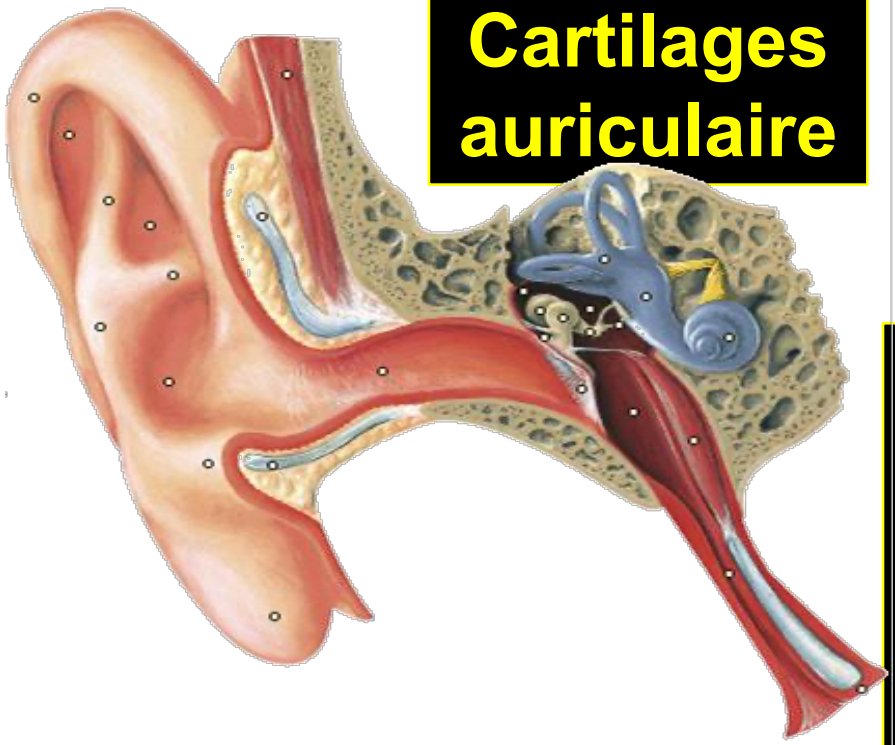
Labrum acétabulaire



Vue latérale de l'os coxal

(3) Le cartilage élastique

Jaunâtre
Riche en **fibres élastiques**



Cartilages auriculaire



Epiglotte et les petits cartilages du larynx

Partie osseuse

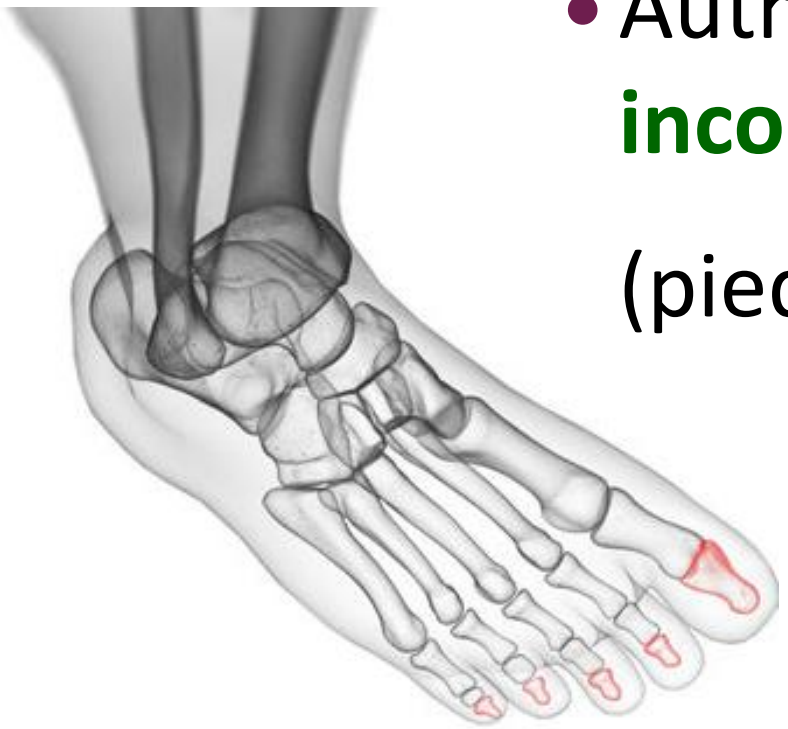




Définition

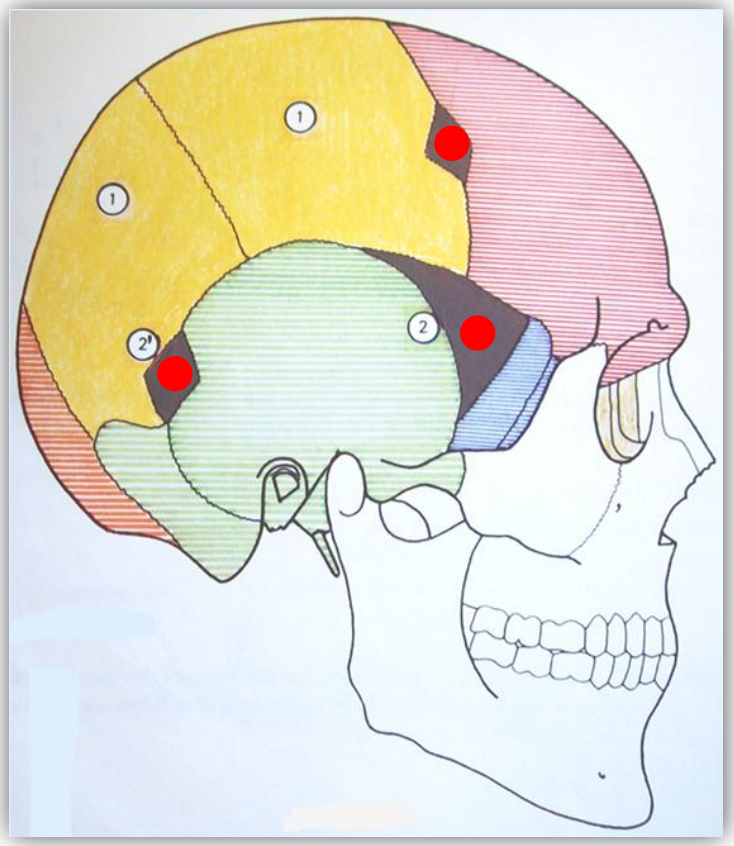
- Les **os** sont **structures rigides et dures** du corps humain.
- **206 os constants**
- Autres **os surnuméraires** ou **inconstants**

(pied, main et crâne)

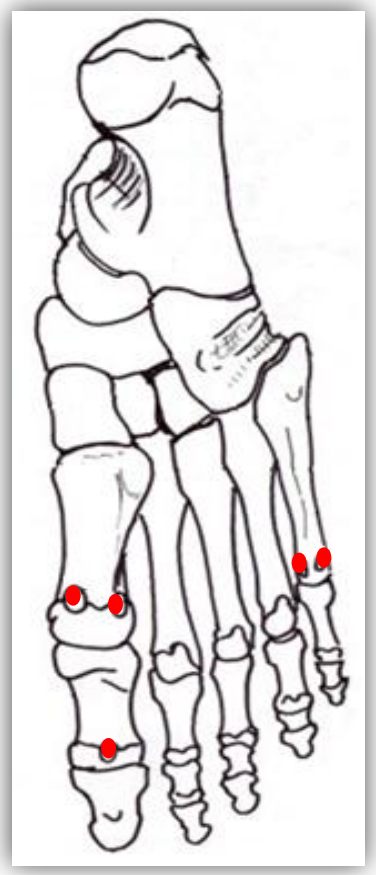


Os surnuméraires

Os wormiens
(crâne)

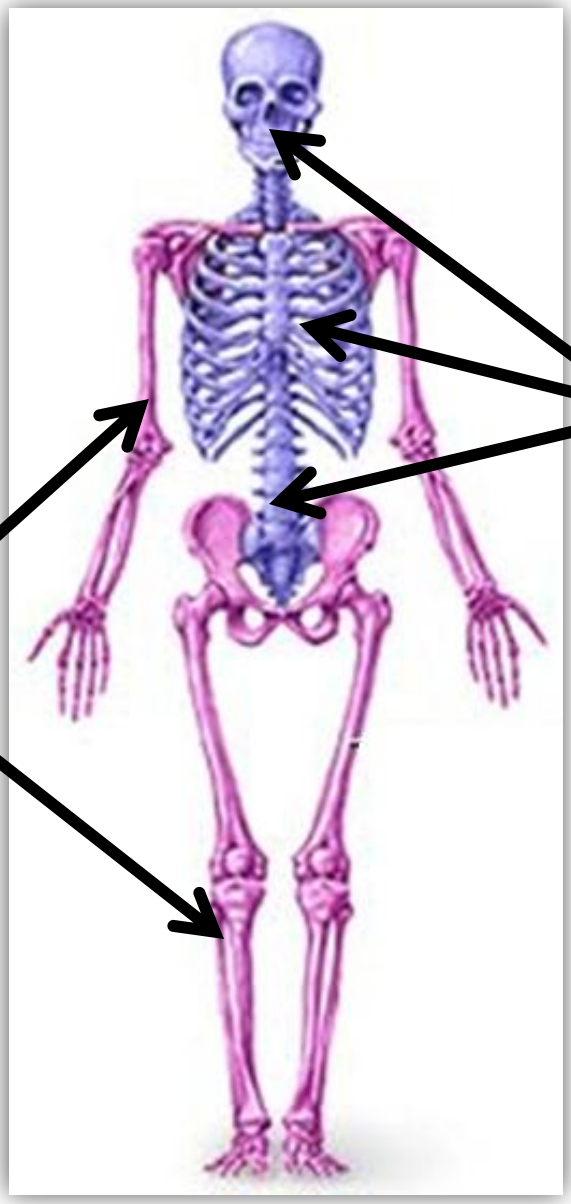


Os sésamoïdes
(pied)




constitution

Constitué de 2 squelettes :



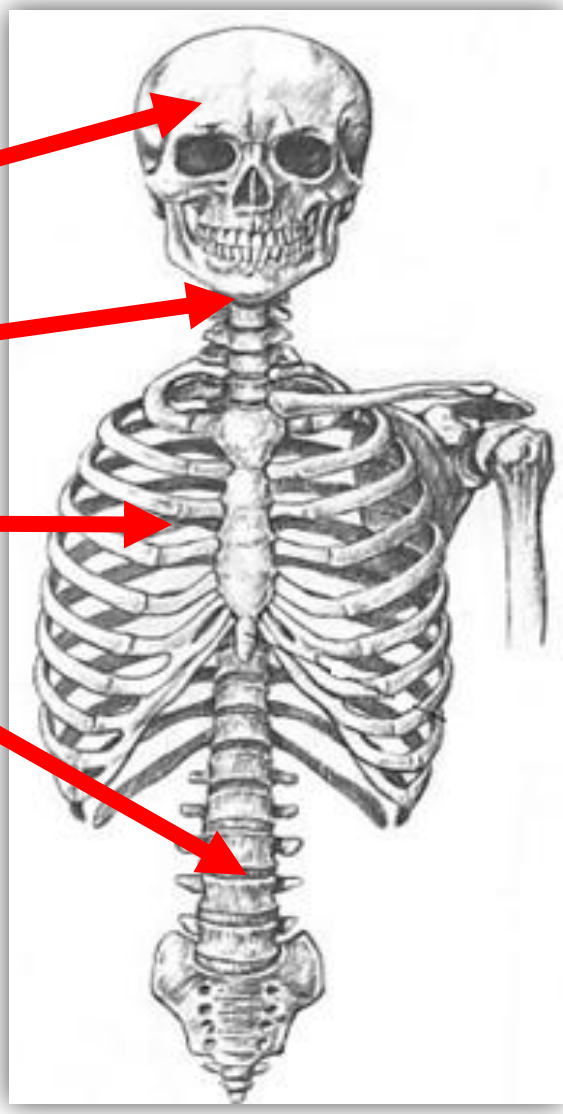
Squelette axial

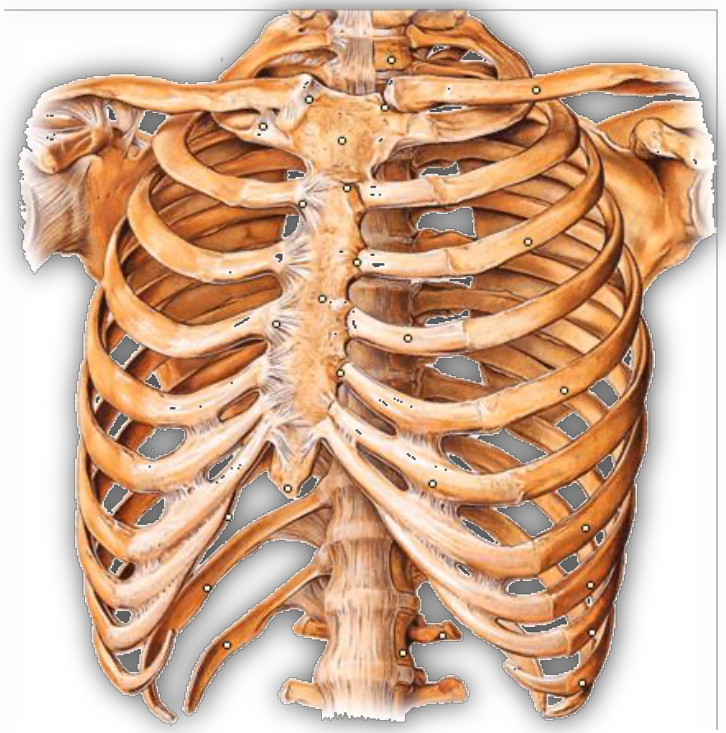
Squelette appendiculaire

(1) Squelette axial

Formé par:

- **Tête osseuse**
- **Os hyoïde**
- **Cage thoracique**
- **Colonne vertébrale (Rachis)**

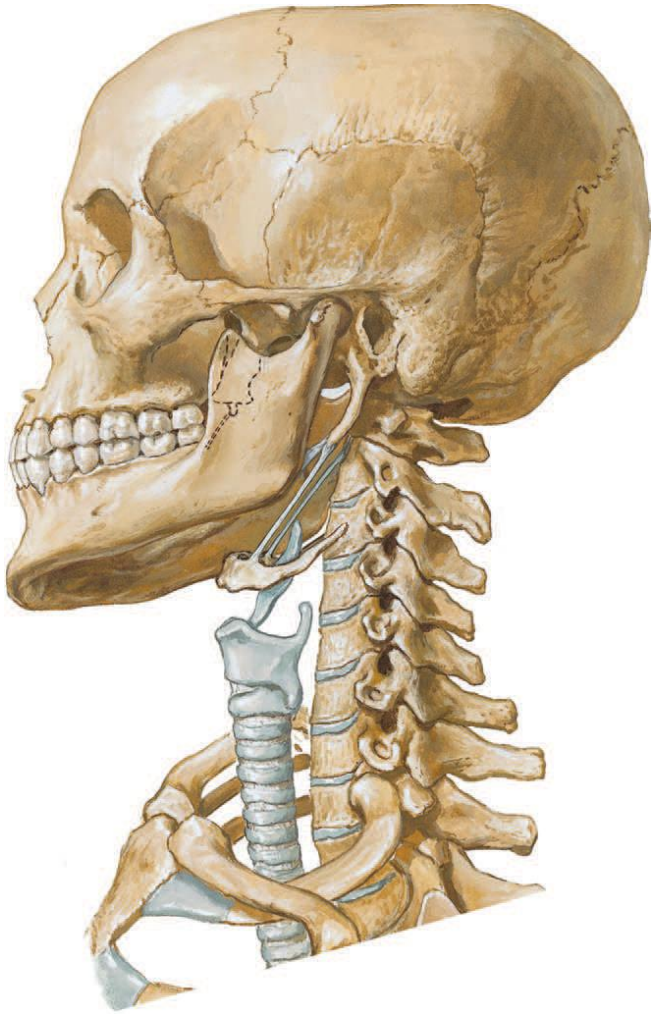




La cage thoracique



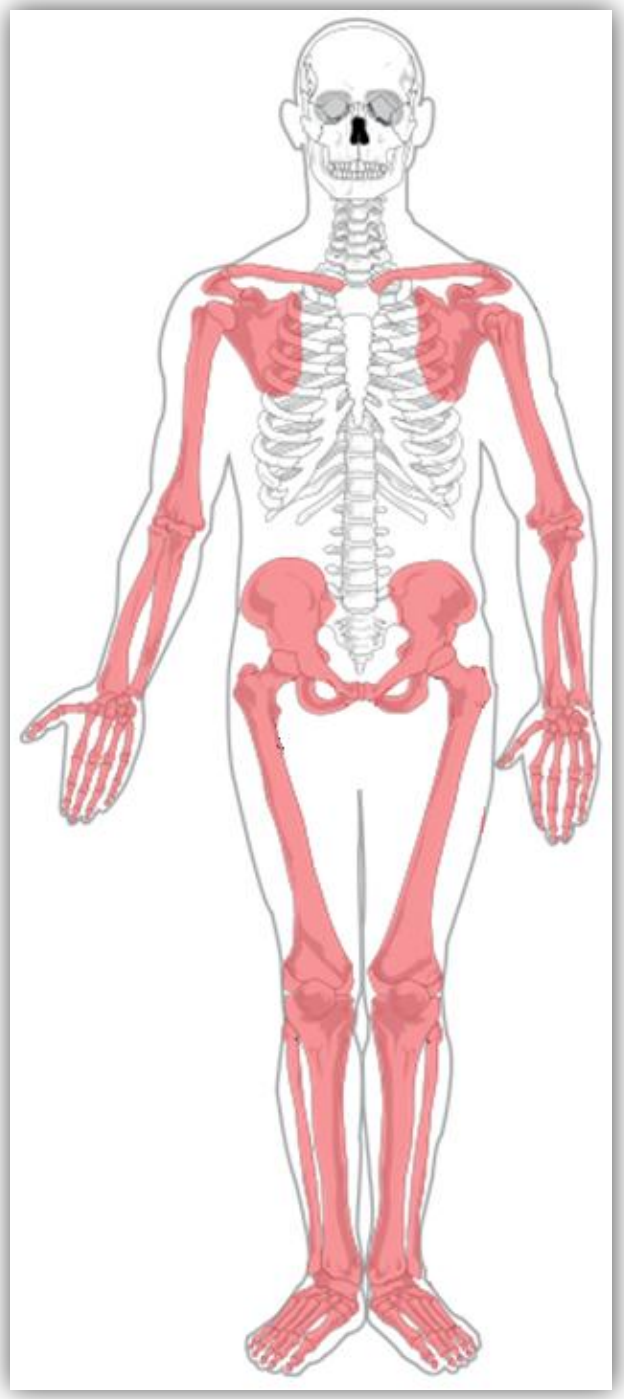
La colonne vertébrale ou rachis



La tête osseuse et **os hyoïde**

(2) Squelette appendiculaire

- **Symétrique**
- **Se fixe** sur squelette axial
- Formé par les **os membres thoraciques** et les os du membre **pelviens**



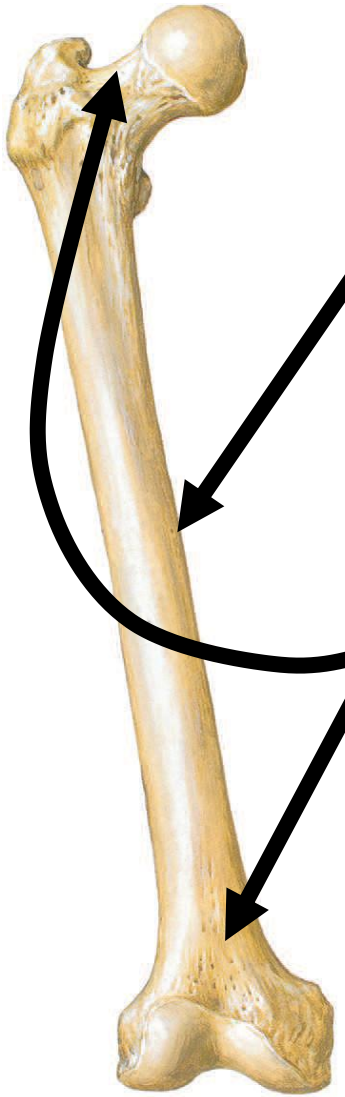


Classification

Il existe 3 types des os

(1) Os long

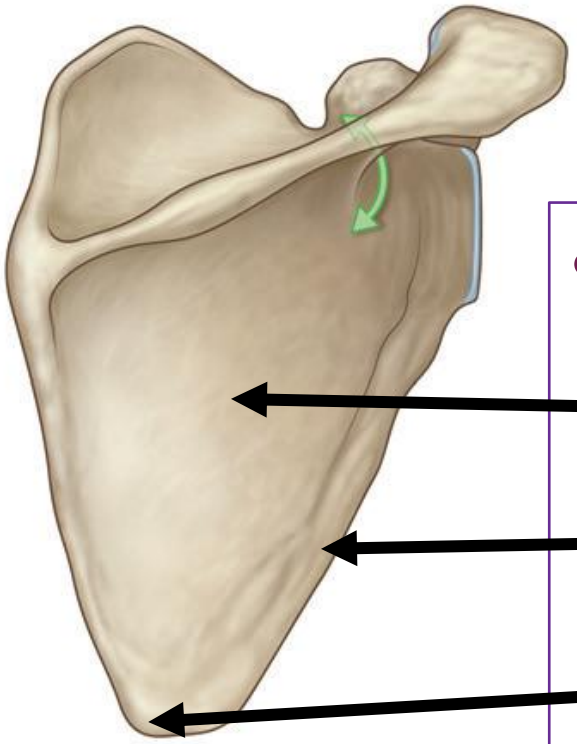
- La **longueur prédomine** par rapport à **la largeur** et l'**épaisseur**
- Exp: **fémur**



- La partie intermédiaire: la **diaphyse (corps)**
- 2 parties distales les **épiphyses** (portent: surfaces articulaires)

(2) Os plat

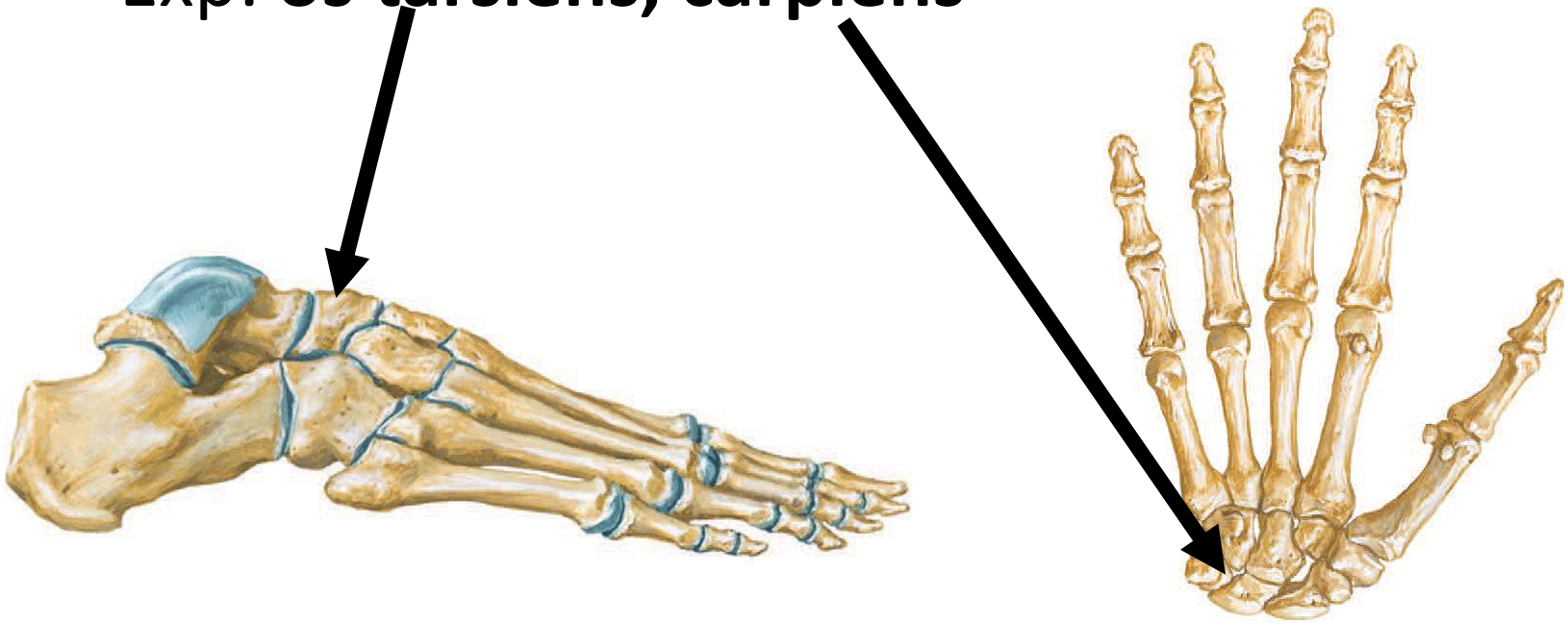
- la **longueur** et la **largeur** **prédominent** par rapport à **l'épaisseur**
- Exp: **scapula**



- Présente:
  - **2 faces**
  - **des bords**
  - **Des angles**

(3) Os court

- **Les 3 dimensions** presque égales
- Exp: **os tarsiens, carpiens**



Remarque

- **L’os irrégulier**

  échappe à la description des 3 types précédents

- Exp: **vertèbre**





Morphologie

- La surface des os **n’est pas régulière**
- Des **reliefs** (éminences, saillies) et des **cavités.**

(1)
Éminences articulaires



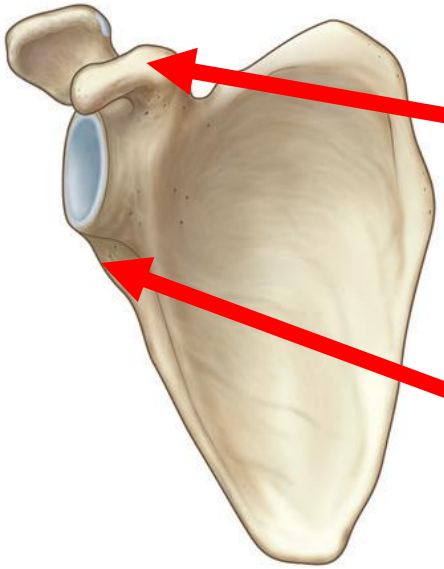
- Ou la **tête**
- Exp : la **tête humérale**,

la **tête fémorale**



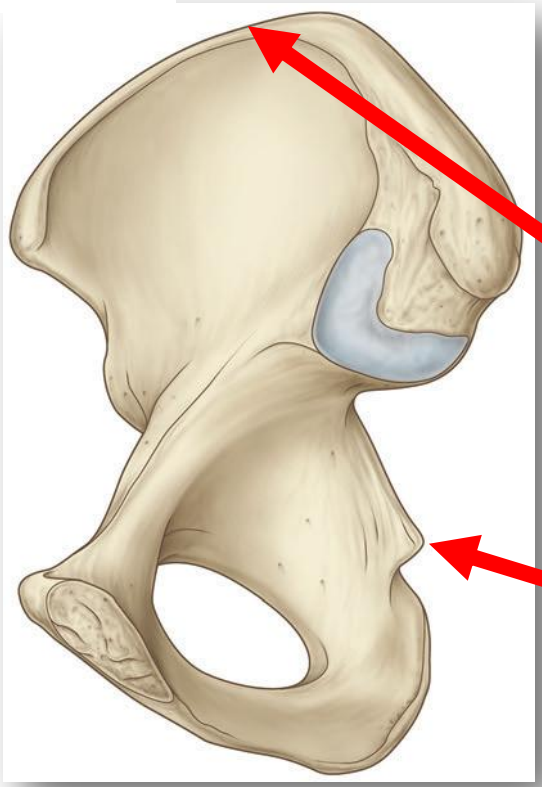
(2)
Éminences non articulaires

- Servent aux:
  attaches ligamentaires
  ou **insertions musculaires**
- Plusieurs **exemples**:



Le processus (apophyse):
éminence volumineuse de forme variable
(**processus coracoïde**)

Le tubercule :
petite éminence arrondie
(**tubercule infra-glénoïdien**)

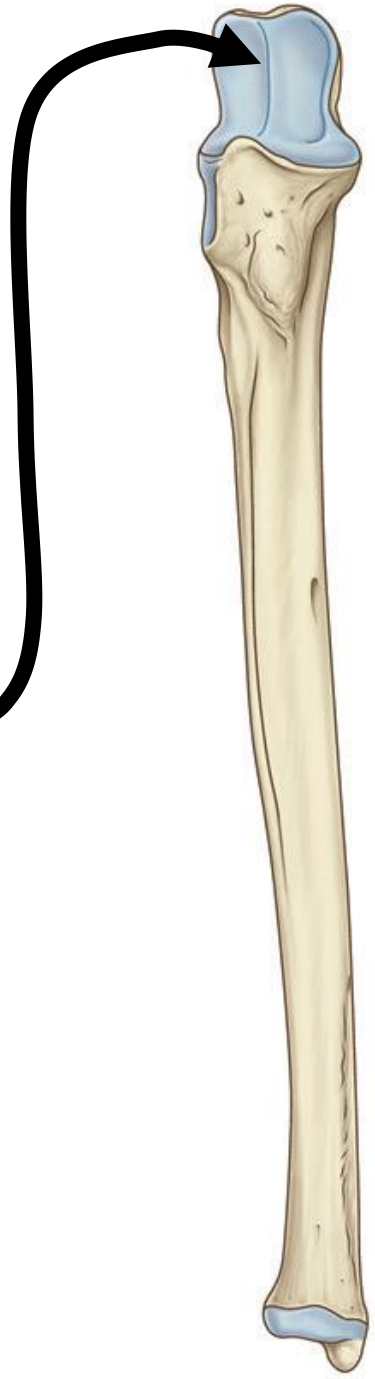


La crête :
éminence linéaire
(**crête iliaque**)

L'épine :
éminence aiguë
(**épine sciatique**)

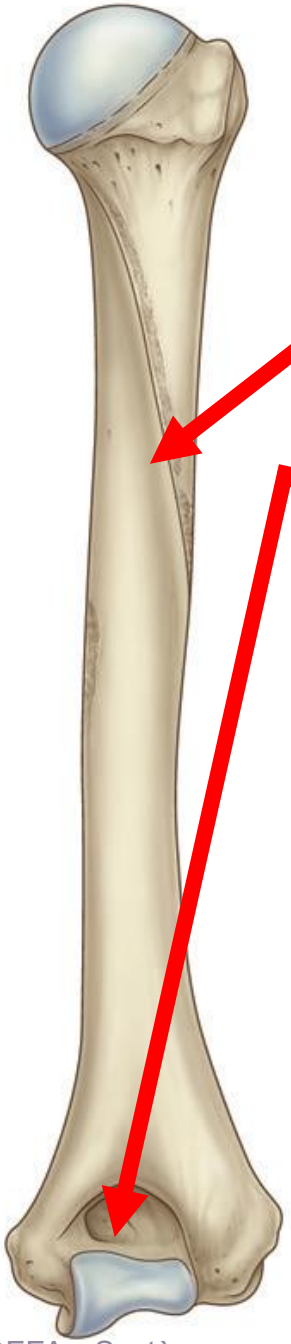
(3)
Cavités articulaires

- Répondent aux **éminences articulaires**
- Exp :
  incisure trochléaire de l'ulna



(4)
Cavités
non articulaires

- Servent aux:
  passage ou **attache** **d'un élément anatomique**
- Plusieurs **exemples**:

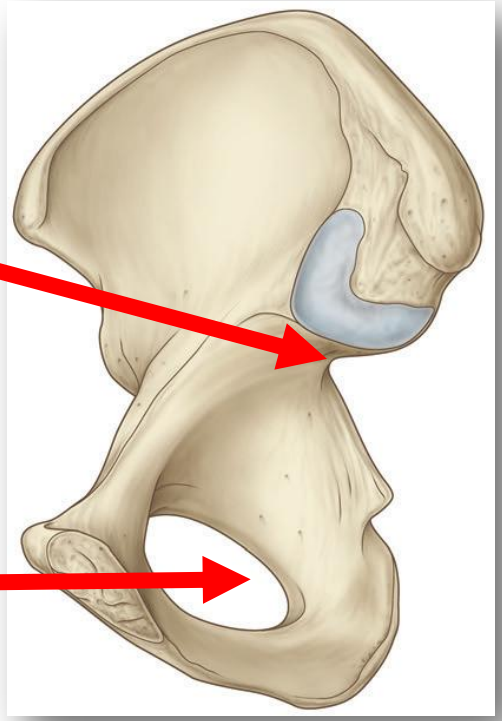


Le sillon (gouttière) :
canal ouvert
(sillon **radial**)

La fossette :
creux plus au moins large et profond
(**fossette olécranienne**)

L'incisure (échancrure) :
Cavité creusée au niveau d'un bord
(**incisure sciatique**)

Le foramen (trou) :
Ouverture dans l'os
(le foramen obturé)

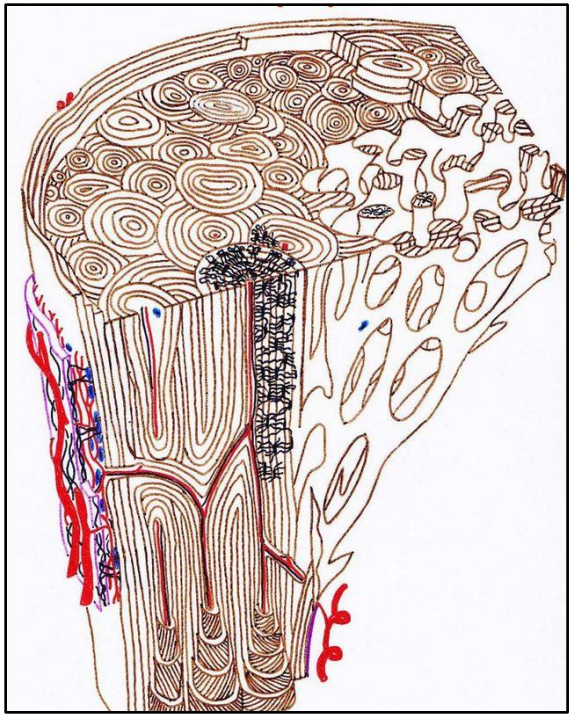


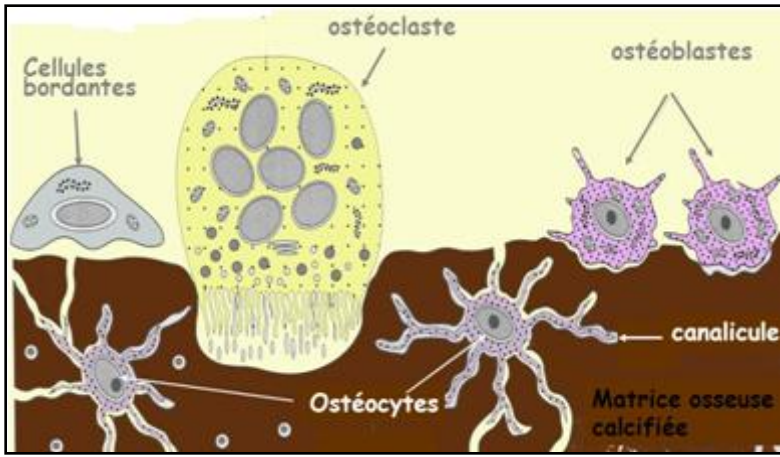


Structure

L'os est un tissu conjonctif spécialisé, dense et solide = **tissu osseux**

- Il est :
  - Riche en **phosphate** et **carbonate de calcium**
  - Richement **vascularisé**
  - En **perpétuel remaniement**
  - Formé par **3 types** de **cellules osseuses** comprises dans une **matrice intercellulaire calcifiée**



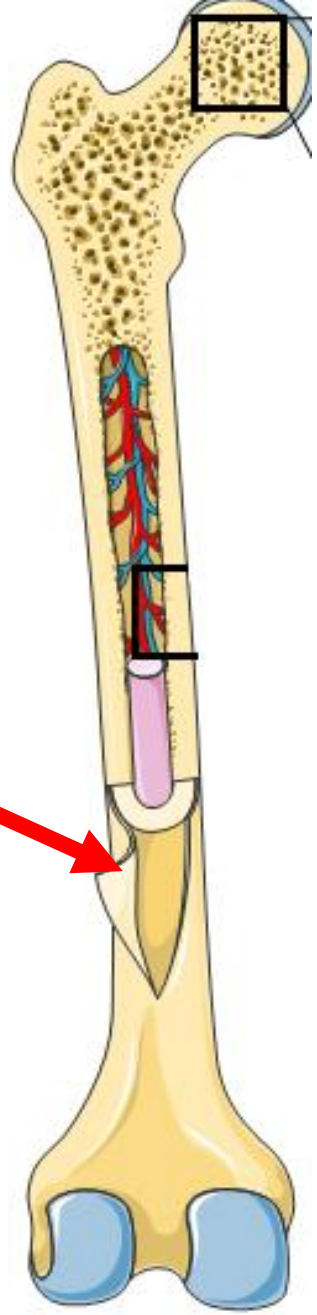


L’os est constitué par 4 éléments:

- ❑ **Le périoste**
- ❑ **L’os compact**
- ❑ **L’os spongieux**
- ❑ **Le canal médullaire**

(1) Périoste

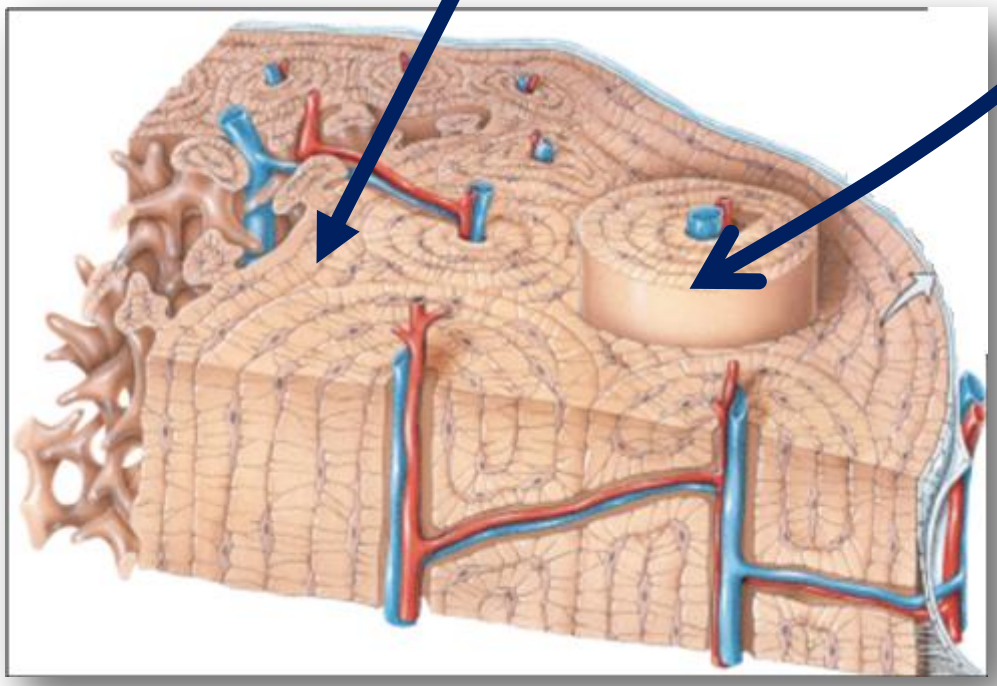
- **Le périoste est une membrane blanchâtre**
- **Recouvre l'os** sauf au niveau des **surfaces articulaires**

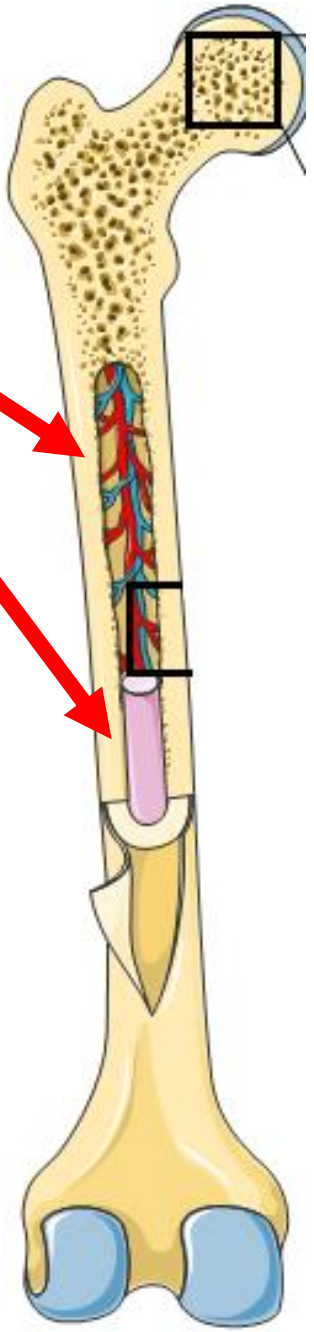


(2) Os compact

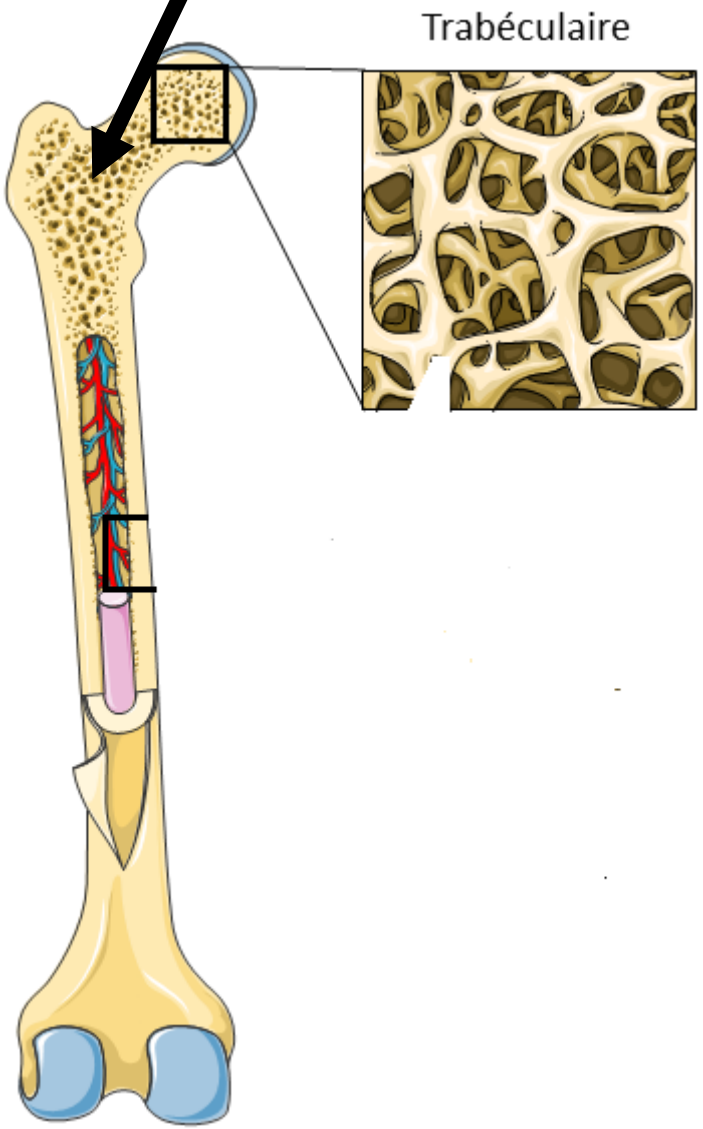
- **Substance dure** et **blanche**
- Formé d'**ostéons**

  et **lamelles interstitielles**



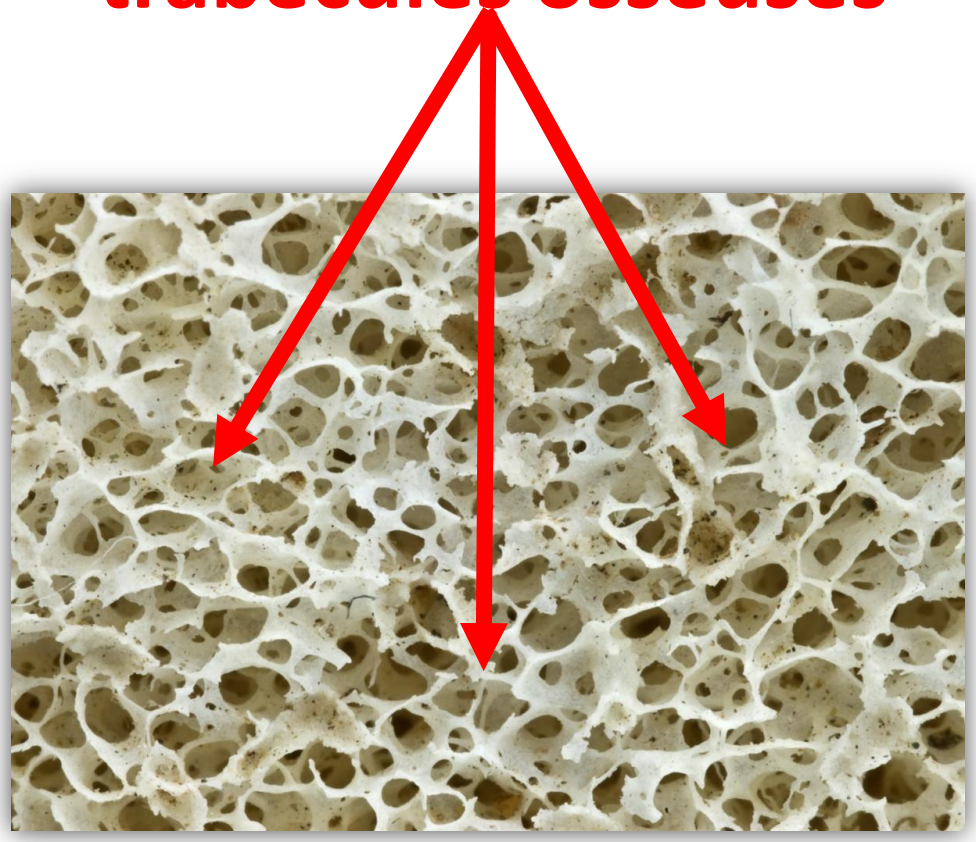


(3) Os spongieux



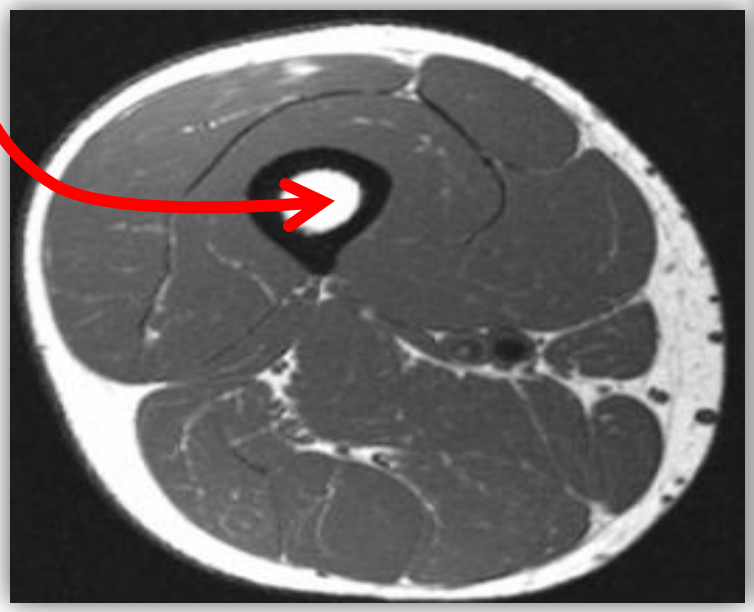
- Constitué par des

trabécules osseuses

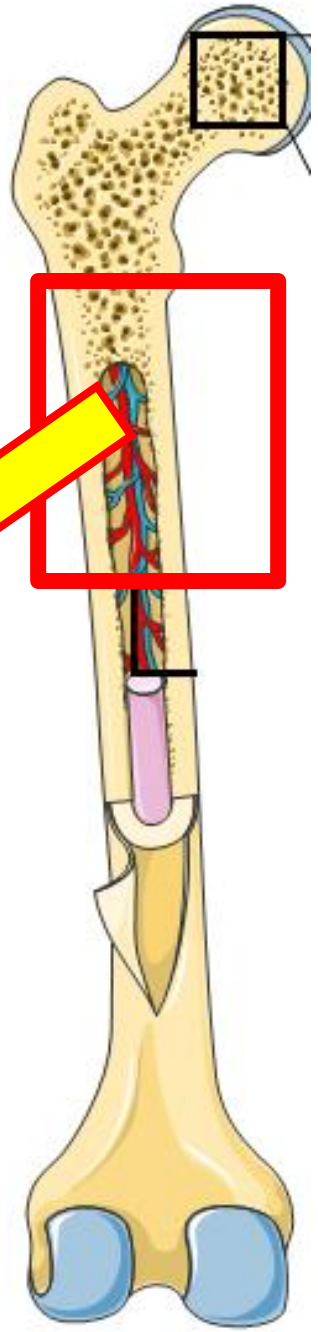


(4) Canal médullaire

- Creusé dans l'os
- Contient:
  - **Moelle osseuse**
  - **Vaisseaux**
  - **Nerfs**



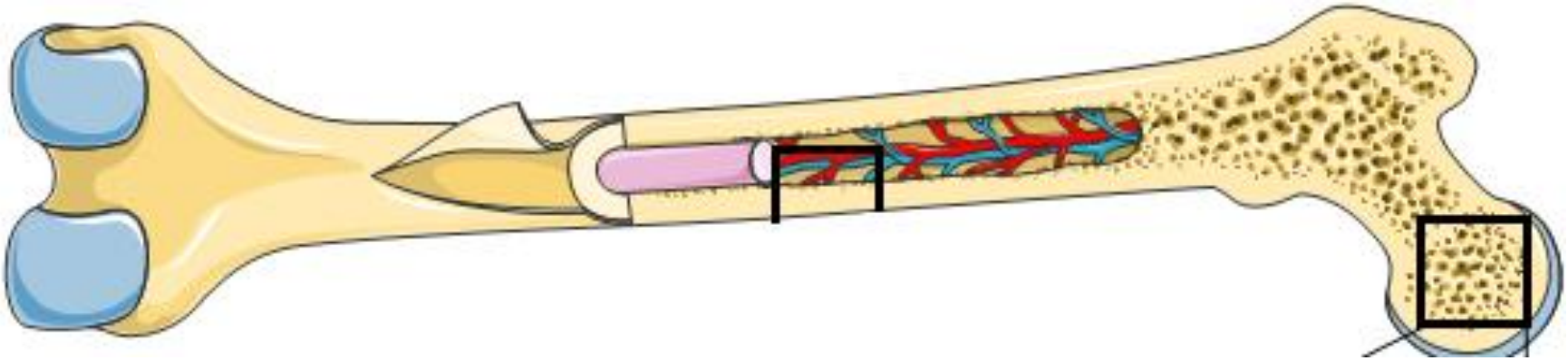




L’os compact et l’os spongieux se disposent différemment selon les types des os

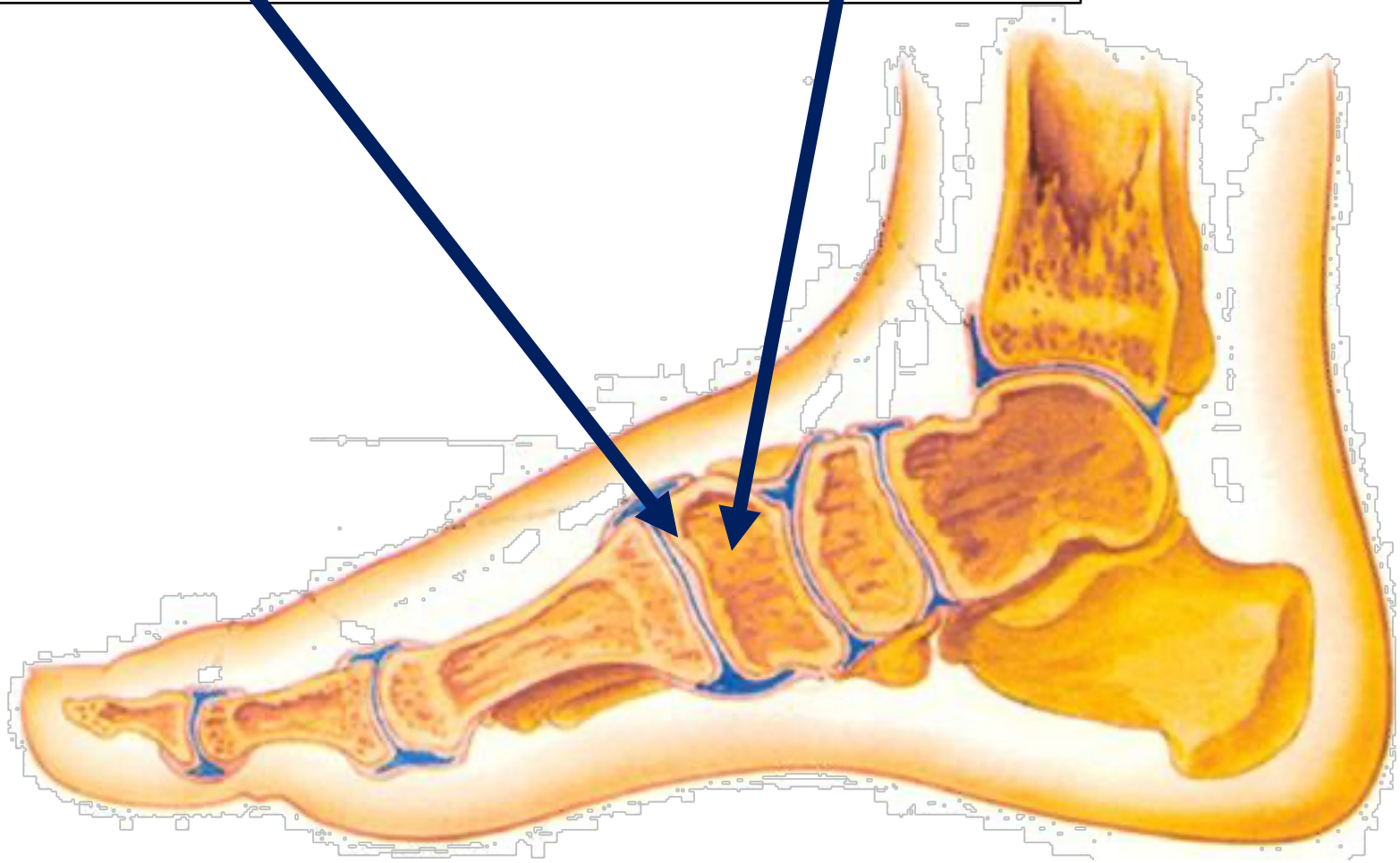
Os long :

- **Os compact** diaphysaire
- **Canal médullaire** diaphysaire
  (contient moelle osseuse jaune)
- **Os spongieux** épiphysaire
  (contient moelle osseuse rouge)



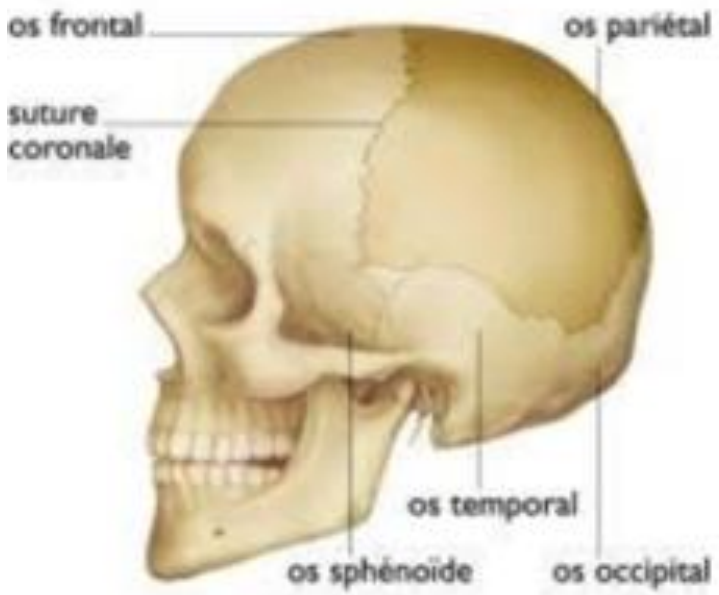
Os court:

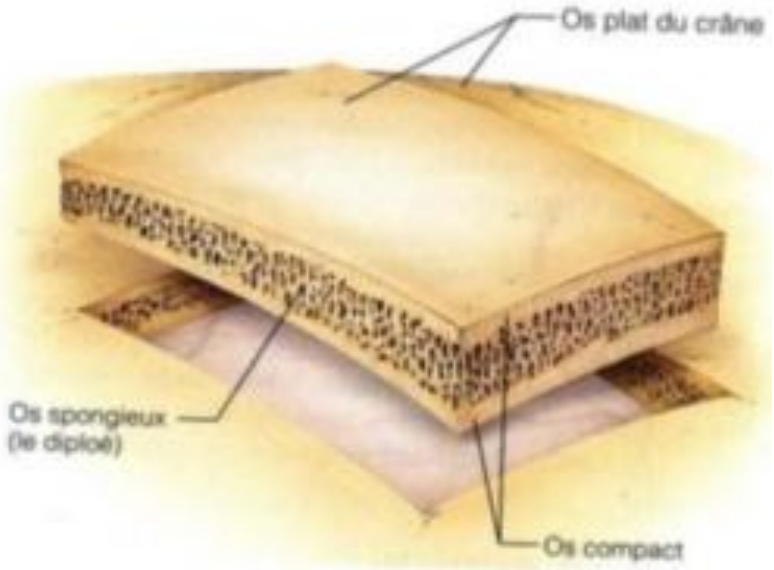
– **Os compact** entourant **os spongieux**



Os plat :

- **Os spongieux** (diploé), en couche moyenne
- **Os compact** (vitrée), en couches externe et interne

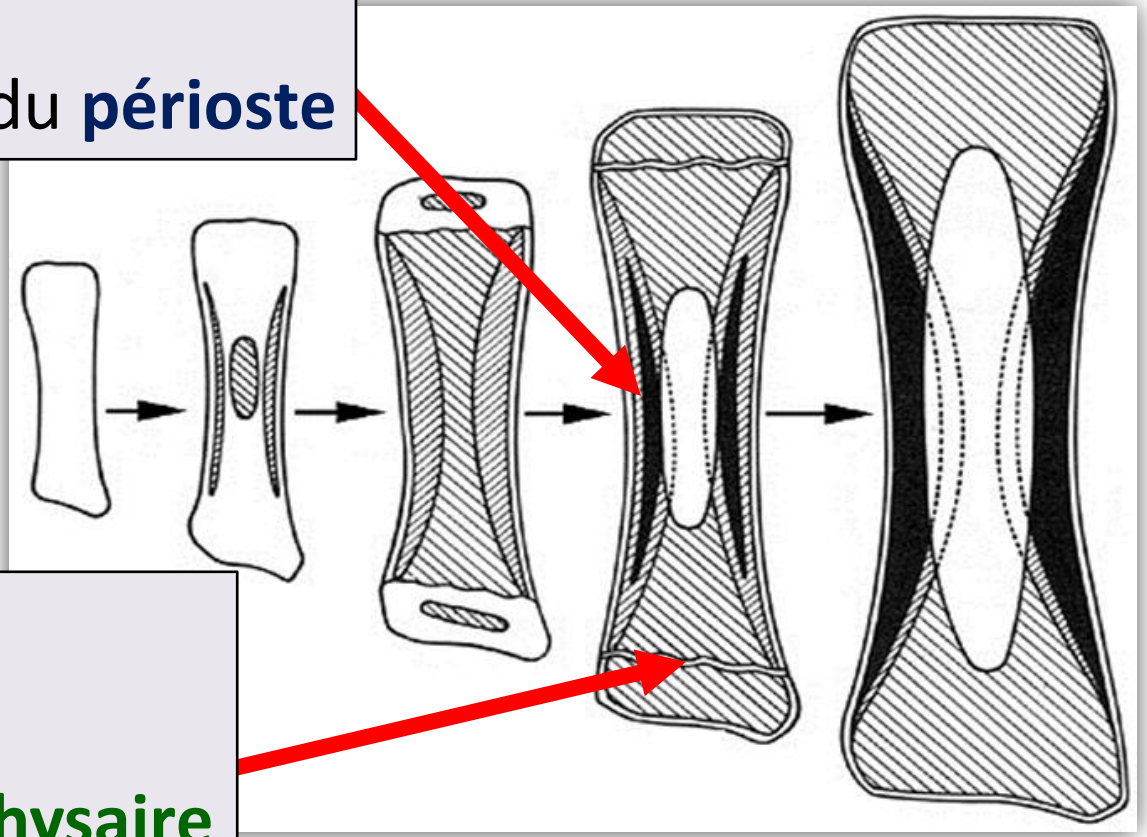





Croissance

2 types de croissance

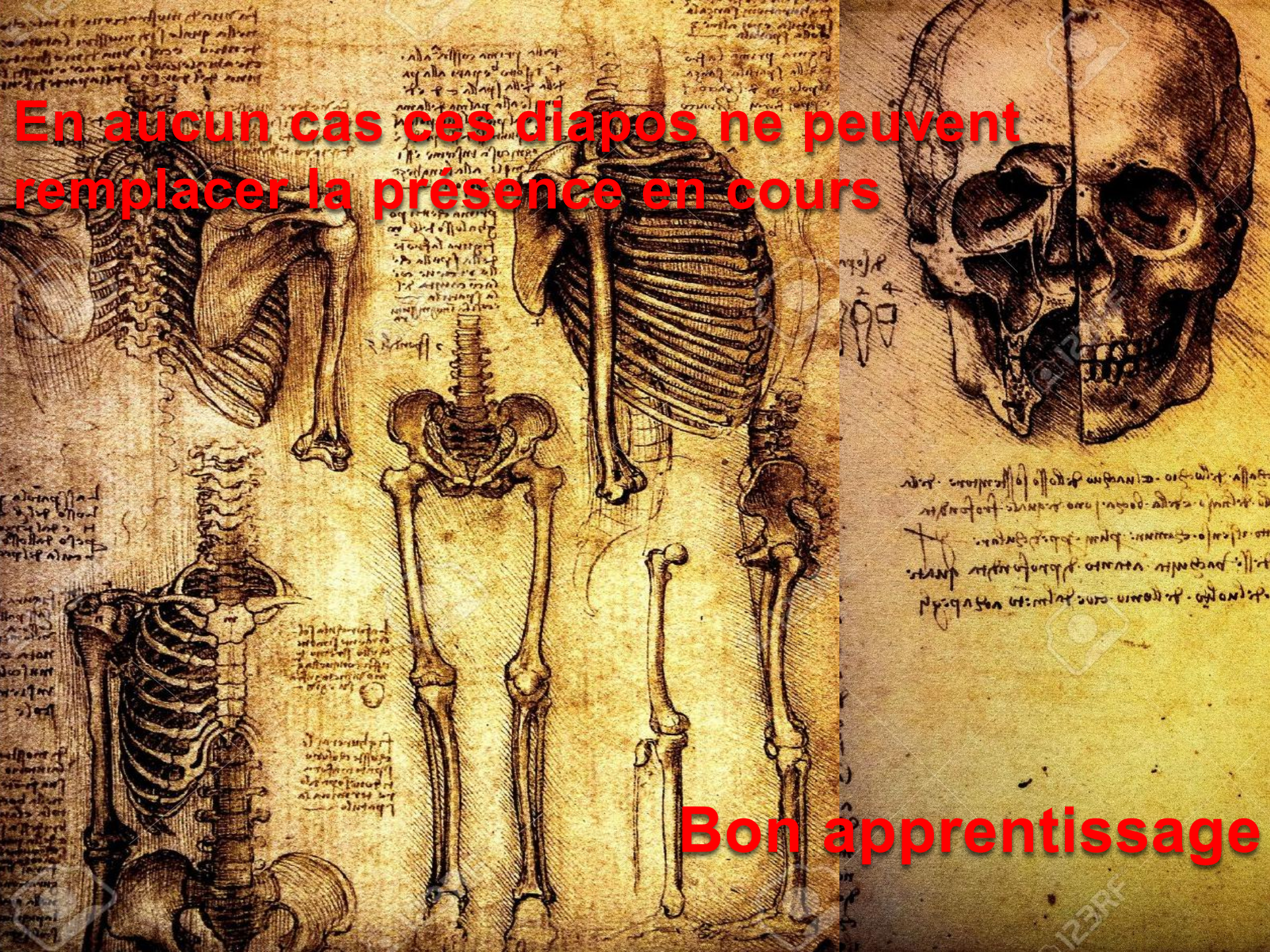
En largeur:
se fait au niveau du **périoste**



En longueur:
se fait au niveau
du **cartilage épiphysaire**
(de conjugaison)



Cartilage de conjugaison (flèches)


En aucun cas ces diapos ne peuvent remplacer la présence en cours
Bon apprentissage